

$$\begin{aligned}
\frac{\partial}{\partial v_c} \log(p(o|c)) &= u_o - \frac{1}{\sum_{w=1}^V \exp(u_w^T v_c)} \cdot \left(\sum_{x=1}^V \exp(u_x^T v_c) u_x \right) \\
&= u_o - \sum_{x=1}^V \frac{\exp(u_x^T v_c)}{\sum_{w=1}^V \exp(u_w^T v_c)} u_x \quad \text{Distribute term across sum} \\
&= u_o - \underbrace{\sum_{x=1}^V p(x|c) u_x}_{\text{This an expectation: average over all context vectors weighted by their probability}} \\
&= \text{observed} - \text{expected}
\end{aligned}$$

This is just the derivatives for the center vector parameters
 Also need derivatives for output vector parameters
 (they're similar)
 Then we have derivative w.r.t. all parameters and can minimize